Nanda Kumar

India nanduaru333@gmail.com 8971399165 in/nandakumar-arjun

SUMMARY

Full Stack Engineer with 2+ years of experience building AI-powered, real-time monitoring platforms using Python, Django, React, and PostgreSQL. Specialized in low-latency dashboards, rule-based alerting, and ML-integrated systems. Led CV and safety analytics at Seewise.AI. Skilled in Docker, Prometheus, and scalable backend design for time-series data and high-performance industrial solutions.

SKILLS

Backend: Django, Django REST Framework, Flask, RESTful APIs, Node.js (basic), JWT Authentication, Role-Based Access Control (RBAC), WebSocket Integration, Asynchronous Processing

Frontend: React.js (Hooks, Context API, React Router), JavaScript (ES6+), HTML5, CSS3, Chart.js, ApexCharts, Material-UI, Bootstrap, WebSockets, Responsive UI Design

Database: PostgreSQL, MySQL, SQLAlchemy ORM, Query Optimization, Time-Series Data Modeling, JSON Fields

AI/ML Integration: Computer Vision (CV model integration), Image Annotation Platforms, Model Training Workflow Integration, Real-Time Inference APIs

DevOps & Tools: Docker, GitHub Actions (CI/CD), Prometheus, Grafana, Git, Linux, Nginx (basic)

General: Agile/Scrum Methodology, MVC & Clean Architecture, API Versioning & Error Handling, PDF/CSV Exporting, Token-Based Auth Flows

Languages: Python, JavaScript, SQL

EXPERIENCE

Software Development Engineer I

Seewise.AI — **February 2023 - Present, Chennai**

- Spearheaded development of **real-time safety and traffic monitoring dashboards** using Django REST Framework + React, with **JWT-secured RBAC APIs** and **live WebSocket visualizations**.
- Engineered **AI-first backend architectures** with asynchronous processing, Redis caching, and Celery queues for high-frequency CV event ingestion and ML inference triggers.
- Designed **low-latency dashboard pipelines** (e.g., galvanizing, furnace) using **pre-aggregated PostgreSQL cache tables**, scheduler-driven processing (APScheduler), and filter-optimized APIs — enabling **<1s dashboard loads** at scale.
- Built configurable **rule-based alert engines** for L1/L2 violation escalations with image capture, real-time streaming, and historical event tracking.
- Developed an **end-to-end annotation and ML training platform** with concurrent image labeling, dataset versioning, and auto-retraining workflows using Python, React, and Redis task queues.
- Integrated **Prometheus/Grafana dashboards** for deep observability of system memory, CPU, and task-level metrics; reduced system debugging time by over **60%**.
- Automated deployment with **Docker + GitHub Actions**, improving release stability and accelerating feature delivery cycles.
- Contributed to 5+ AI-driven products across **industrial safety, traffic intelligence, annotation tooling, and predictive maintenance** — directly impacting plant safety KPIs and operational insights.
- Launched user friendly UI components and workflow enhancements in React across 5+ AI-driven products over 14 months, conducting monthly A/B
- Directed end-to-end Full Stack Development of five AI-driven products over a 14-month period,

PROJECT

Galvanizing Process Monitoring System

Seewise.AI

- Built real-time dashboards using **Django, Django REST Framework, and ApexCharts**, improving visibility across **3 production shifts by 40%**.
- Engineered `process_api_data` caching with **APScheduler**, reducing dashboard API latency from **6s to under 1s**.
- Optimized **PostgreSQL** with advanced indexing and pooling to handle **10M+ time-series rows** efficiently.
- Enabled multi-format report exports (PDF, CSV, PNG) and custom filters using **HTML-to-PDF libraries** and **Bootstrap UI**.**Tech Stack**: Django, DRF, PostgreSQL, APScheduler, ApexCharts, Bootstrap, HTML-to-PDF.

AI-Powered Safety Monitoring System

Seewise.AI

- Developed a real-time safety monitoring system using **React + Django REST Framework**, reducing manual audits by **60%** across **200+ camera zones**.
- Integrated **WebSockets + Redis** to stream alerts with **<500ms latency** and **98% violation image accuracy**.
- Built flexible, role-aware alert rules using **custom RBAC middleware**, enabling **10+ site-specific safety policies**.
- Applied lazy loading and async-safe API patterns to boost UI load performance by **35%**.**Tech Stack**: React.js, Django REST Framework, Redis, WebSocket, PostgreSQL, OpenCV, Role-Based Access Control.

Image Annotation & ML Training Platform

Seewise.AI

- Developed an interactive annotation interface using **React, Flask, and SQLAlchemy**, enabling **100K+ image labels** synced with live camera streams.
- Scaled background job throughput by **4x** using **Celery and Redis**, supporting real-time annotation and model training pipelines.
- Parallelized training workflows using **Python threading**, reducing ML cycle time from **3 days to under 6 hours**.
- Implemented dataset versioning, improving traceability and reducing model errors by **50%**.**Tech Stack**: React.js, Flask, Celery, Redis, Python Threading, SQLAlchemy, HTML5/CSS3.

Real-Time Traffic Monitoring & Analytics System

Seewise.AI

- Built a live dashboard using **React and Flask** to monitor **100+ CCTV feeds**, tracking congestion, people, and vehicles in real time.
- Enabled streaming of traffic metrics using **WebSocket and Redis pub/sub**, maintaining **<1 second update delay** for **50+ concurrent users**.
- Handled **10K+ events/hour** using multi-threaded endpoints and non-blocking I/O to maintain high throughput.
- Integrated **Prometheus and Grafana** for system observability, reducing debugging time by **50%**.**Tech Stack**: React.js, Flask, Redis, WebSocket, Chart.js, Prometheus, Grafana, Docker.

Furnace Monitoring & Analytics Platform

Seewise.AI

- Developed dashboards using **Django Templates and ApexCharts** to monitor **5+ industrial furnaces** with real-time temperature tracking.
- Integrated **PLC/IoT sensor data** via WebSocket APIs, ingesting **50K+ events/day** with sub-second anomaly detection.
- Automated alerts and shift-wise emails for **100+ monthly threshold violations**, reducing delayed incident response by **60%**.
- Designed archiving and retention logic for **6+ months of time-series furnace data**, ensuring performance and compliance.**Tech Stack**: Django, PostgreSQL, WebSocket, ApexCharts, Cron Jobs, IoT Integration, Email Alerts.

EDUCATION

MBA

Bangalore University • Bangalore • 2021 • 73.5